

kNN (k=3)

You are working as a data scientist in a fruit company. You would like to develop a model to judge whether a given fruit is orange or grapefruit.

You have a dataset in CSV format, which is stored in the “fruit_dataset.csv” (see reading materials). The file contains 6 columns:

- name: class labels, can be {orange, grapefruit}
- diameter: diameter of the fruit in centimeters
- weight: weight of the fruit in grams
- red: average red reading from an RGB scan
- green: average green reading from an RGB scan
- blue: average blue reading from an RGB scan

The dataset has been split into predictors (X) and labels (Y).

Train a kNN model with k = 3 to classify the types of fruits.

Script

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 MATLAB Documentation (https://www.mathworks.com/help/)

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```
1 % Load data
2 data = readtable('fruit_dataset.csv');
3 X = data(:, 2:6);
4 Y = data(:, 'name');
5 % Data split
6 Train_X = X(1:7500,:);
7 Train_Y = Y(1:7500,:);
8 Test_X = X(7501:10000,:);
9 Test_Y = Y(7501:10000,:);
10
11 % KNN model with k=3
12 k = 3;
13 model_1 = fitcknn(Train_X, Train_Y, 'NumNeighbors', k); % complete this line
14
15
```

 Run Script



Tests: All Tests Passed

Submit



	Is fitcknn present?
	Is the KNN model identical to the reference?
	Prediction testing

Output

Code ran without output.